



HOUSSEM JARRAY

Embedded Software Engineer (ICT)

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🌐 houssemjarray

👤 Houssem-Jarray

Embedded Systems Engineer, holding a degree in Industrial Computing from ENET'COM Sfax and currently a PhD candidate at SUP'COM. Passionate about the design of intelligent and connected systems, with strong expertise in C/C++ programming, STM32, FreeRTOS, and Embedded Linux. Experienced in Yocto integration, device driver development, and C++ backend and Qt application development. Particularly interested in IoT, reconfigurable systems, and embedded intelligence.

PROFESSIONAL EXPERIENCE

• 06/2025 - 10/2025

C.S.F

Ariana, TN

EMBEDDED SYSTEMS DEVELOPMENT

- **Strong experience in Embedded C development:** memory manipulation, pointer management, low-level operations, and driver creation using memory-mapped registers.
- **Proficient in STM32 architecture (ARM Cortex-M):** peripheral configuration (GPIO, UART, ADC, DMA, etc.), use of the HAL library, NVIC, and CubeMX/CubeIDE tools.
- **FreeRTOS :** Skilled in real-time multitasking management, including task scheduling, queues, semaphores, and mutexes.
- **Object-Oriented Programming (C++) and Embedded GUI development with Qt:** Good knowledge of OOP concepts in C++ and development of Qt-based interfaces for embedded systems.
- **Embedded Linux User Space:** Understanding of embedded Linux system architecture, shell scripting, boot process, library compilation and management, toolchain usage, and cross-compilation.
- **Embedded Linux and IoT Programming:** Experience with embedded Linux, from cross-compilation to kernel configuration and driver development on Raspberry Pi, with IoT integration using MQTT.
- **Keywords:** Embedded C, C++, STM32, FreeRTOS, Embedded Linux, Qt, IoT, Debugging.

• 07/2023 - 04/2024

Xpermedia

Sousse, TN

SOFTWARE ENGINEER

- Development of advanced multimedia features in Go, with performance optimization using goroutines and concurrent video decoding management.
- Design of video editing and generation solutions using FFmpeg, creation of REST/GraphQL APIs, and integration of an ORM for database management.
- Implementation of AWS cloud uploads and automation tools for multimedia processing.
- **Keywords:** Go, FFmpeg, REST, GraphQL, AWS, Docker, Git.

• 02/2023 - 06/2023

ZNet-IT

Sfax, TN

PROJET DE FIN D'ÉTUDES

- Development of an intelligent web and mobile DMS (Document Management System) application using Flutter and Node.js, integrating an AI-based OCR model for automatic extraction and classification of digital documents.
- **Keywords:** Flutter, Node.js, React.js, MongoDB, Python, TensorFlow, OCR, Git.

• 07/2022 - 09/2022

Sagemcom

Megrine, TN

ENGINEERING INTERNSHIP

- Development of an automated testing tool to evaluate the functionality of a video player embedded in a digital set-top box.
- Authoring and preparation of technical documentation detailing testing procedures and results.
- **Keywords:** Go, Linux, Embedded Linux, shell scripting, Git, JSON, HLS, DASH.

• 07/2021 - 08/2021

SDI

Kairouan, TN

INTRODUCTORY INTERNSHIP:

- Design and development of a user management application based on CRUD operations, using the Angular framework for the frontend and Node.js for the backend.
- **Keywords:** Angular, Node.js, Express.js, REST API, TypeScript.

SKILLS

- **Programming Languages:** C/C++, Rust, Java, Go
- **Modeling Languages:** UML, VHDL
- **Operating Systems:** Linux, FreeRTOS
- **Strong Knowledge of:** Buildroot, Yocto, RTOS, CMake
- **Science Informatique :** QT, Docker, kubernetes, SQL, Jenkins.
- **Computer Science / Tools:** Raspberry Pi, STM32, FPGA, ESP32, DSP
- **Networking & Protocols:** LAN, WAN, TCP/IP, UDP
- **Version Control Systems:** Git, GitLab, GitHub
- **Bug Tracking Tools:** JIRA, Mantis Bug Tracker
- **Scripting Languages:** Bash, sh, Python
- **Outils et IDEs :** Eclipse , IntelliJ, QT Creator, VS Code.
- **Standards & Protocols:** SPI, I2C, UART, CAN, MODBUS, LIN buses
- **Methodologies:** Agile Scrum, Lean Six Sigma

CERTIFICATIONS

- C Programming for Embedded Systems – CSF
- STM32 Microcontroller Architecture and Programming – CSF
- Object-Oriented Programming (C++) and Qt GUI for Embedded Systems – CSF
- Embedded Linux User Space – CSF
- POSIX Programming – CSF
- Embedded Linux Kernel Space and IoT – CSF
- FreeRTOS Multitasking – CSF
- Docker, Kubernetes, and Git Certificate Course – KodeKloud

EDUCATION

• 2025 – PRESENT

SUP'COM
Ariana, TN

• 2020 - 2023

ENET'Com
Sfax, TN

• 2018 - 2020

IPEIK
Kairouan, TN

• 2015 - 2016

Ibn Arafa Chebika School
Kairouan, TN

PHD IN INFORMATION AND COMMUNICATION TECHNOLOGIES

- Research on IoT, intelligent systems, and AI-assisted reconfigurable embedded architectures.

ENGINEERING DEGREE IN INDUSTRIAL COMPUTER ENGINEERING

- **Specialization:** Intelligent and Interconnected Systems

PREPARATORY CYCLE

- **Track:** Physics and Chemistry (PC)

BACCALAUREATE IN EXPERIMENTAL SCIENCES

- **Track:** Experimental Sciences

PROJECTS

EMBEDDED LINUX AND C++/QT PROJECT

• 2025

- Development of an embedded multimedia application in C++/Qt for Raspberry Pi, integrating FFmpeg and GStreamer for real-time playback and streaming (YouTube, Netflix).
- Created a custom Linux image using Yocto, optimized the kernel, and integrated multimedia libraries.
- Implemented a smooth touchscreen interface, Wi-Fi/Bluetooth connectivity, and full project management using Git/GitHub and CMake.
- **Keywords:** C++, Qt, Yocto, DTB, SPI, GStreamer, FFmpeg, Raspberry Pi, CMake, REST API, OpenGL, Git, GitHub

FINAL YEAR PROJECT

• 2022

- Real-time visualization of data acquired from an STM32 board, displayed on a dynamic web page via Wi-Fi communication between the web server and the STM32 board
- **Keywords:** C++, Qt, Yocto, DTB, SPI, GStreamer, FFmpeg, Raspberry Pi, CMake, REST API, OpenGL, Git, GitHub

LANGUAGES

French ● ● ● ○ ○
English ● ● ● ○ ○
Arabic ● ● ● ● ●

INTERESTS

Technological innovation, intelligent system design, embedded AI, open-source, sports, and cinema.